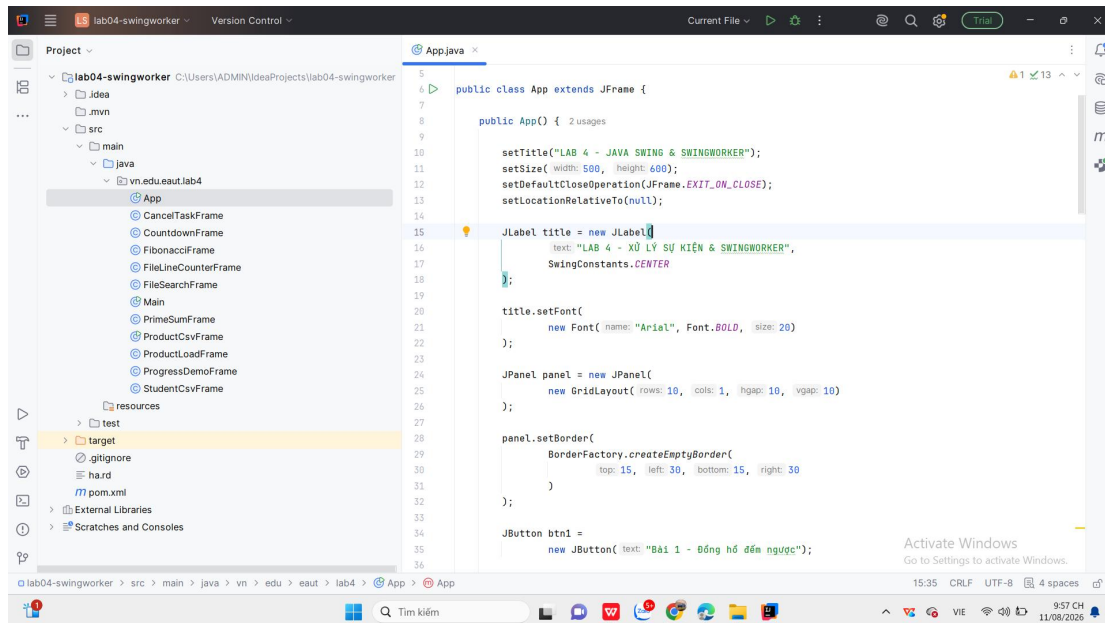


Lab 4: SwingWorker & Multithreading

Họ tên: Trần Việt Đức

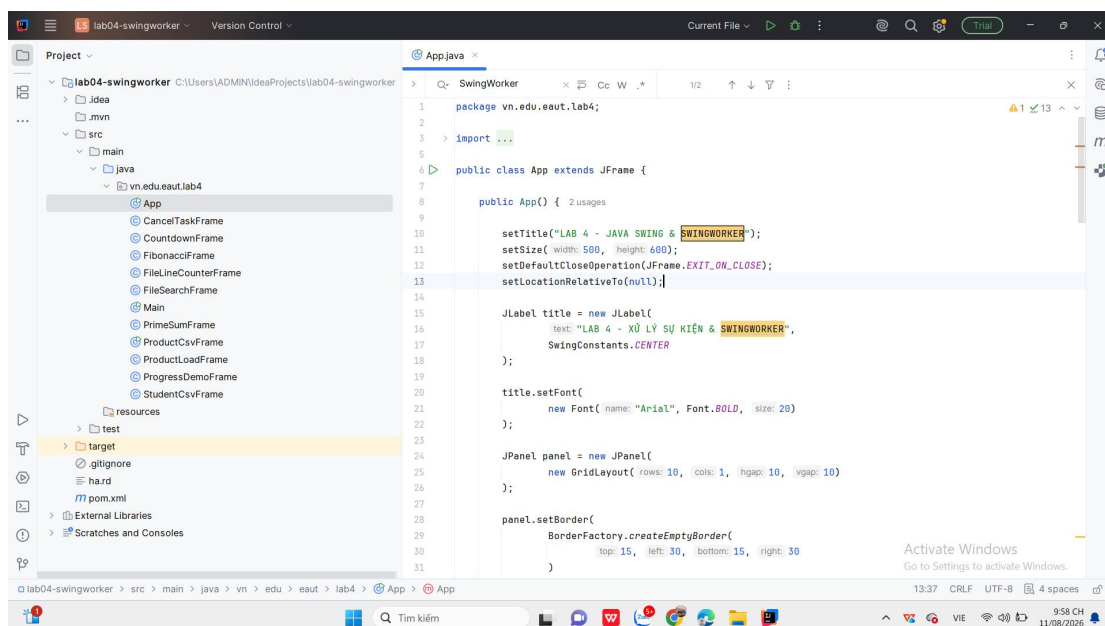
MSV 20220111

Thiết kế project chuẩn theo form

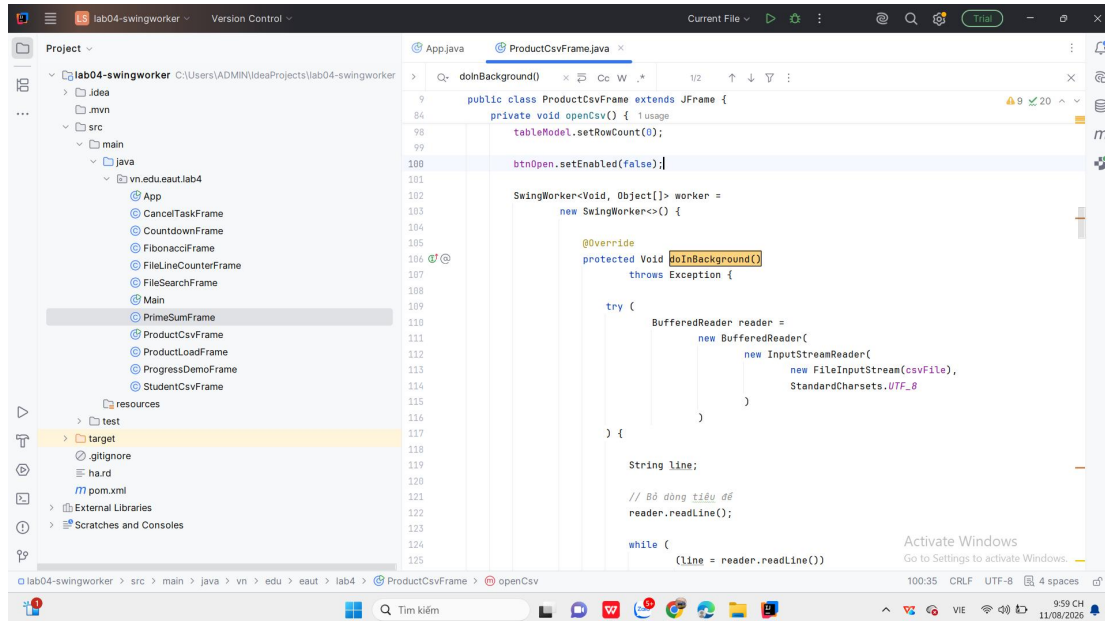


Xử lý logic bất đồng bộ:

Tạo đối tượng SwingWorker cho từng tác vụ.



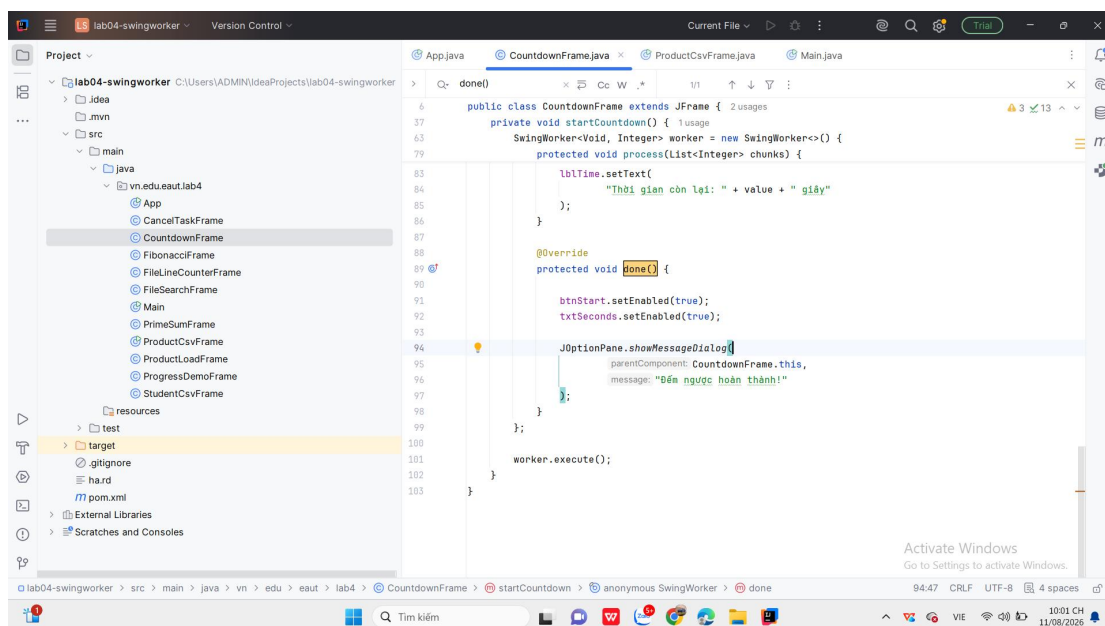
Đưa logic xử lý chính vào doInBackground().



```
public class ProductCsvFrame extends JFrame {  
    private void openCsv() {  
        tableModel.setRowCount(0);  
        btnOpen.setEnabled(false);  
  
        SwingWorker<Void, Object[]> worker =  
            new SwingWorker<>() {  
  
                @Override  
                protected Void doInBackground()  
                    throws Exception {  
  
                    try {  
                        BufferedReader reader =  
                            new BufferedReader(  
                                new InputStreamReader(  
                                    new FileInputStream(csvfile),  
                                    StandardCharsets.UTF_8  
                                )  
                            )  
                    }  
  
                    String line;  
  
                    // Bỏ dòng tiêu đề  
                    reader.readLine();  
  
                    while (  
                        (line = reader.readLine())  
                    ) {  
                        // ...  
                    }  
                }  
            }  
        worker.execute();  
    }  
}
```

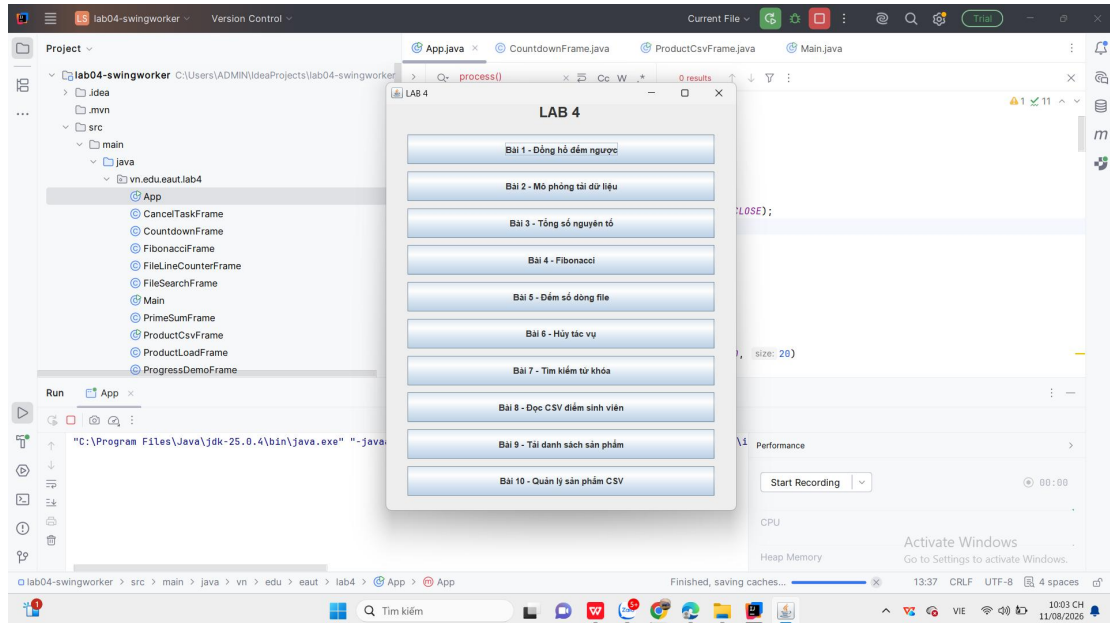
Sử dụng process() để cập nhật các chỉ số trung gian

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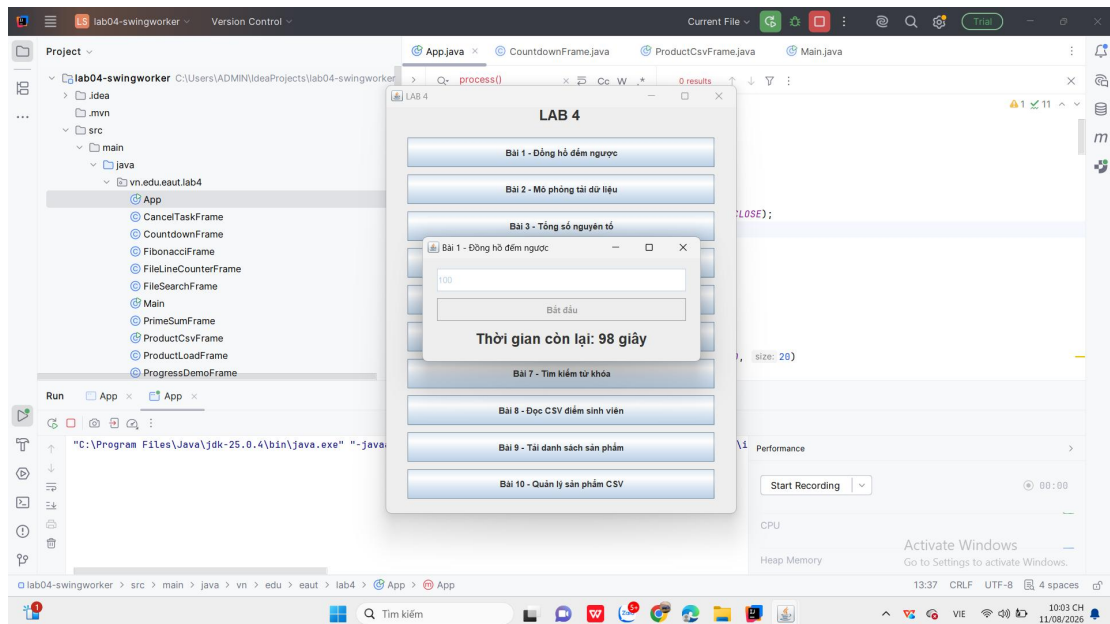


```
public class CountdownFrame extends JFrame {  
    private void startCountdown() {  
        SwingWorker<Void, Integer> worker = new SwingWorker<>() {  
            protected void process(List<Integer> chunks) {  
                lblTime.setText(  
                    "Thời gian còn lại: " + value + " giây"  
                );  
            }  
  
            @Override  
            protected void done() {  
                btnStart.setEnabled(true);  
                txtSeconds.setEnabled(true);  
  
                JOptionPane.showMessageDialog(  
                    parentComponent: CountdownFrame.this,  
                    message: "Đếm ngược hoàn thành!"  
                );  
            }  
        };  
        worker.execute();  
    }  
}
```

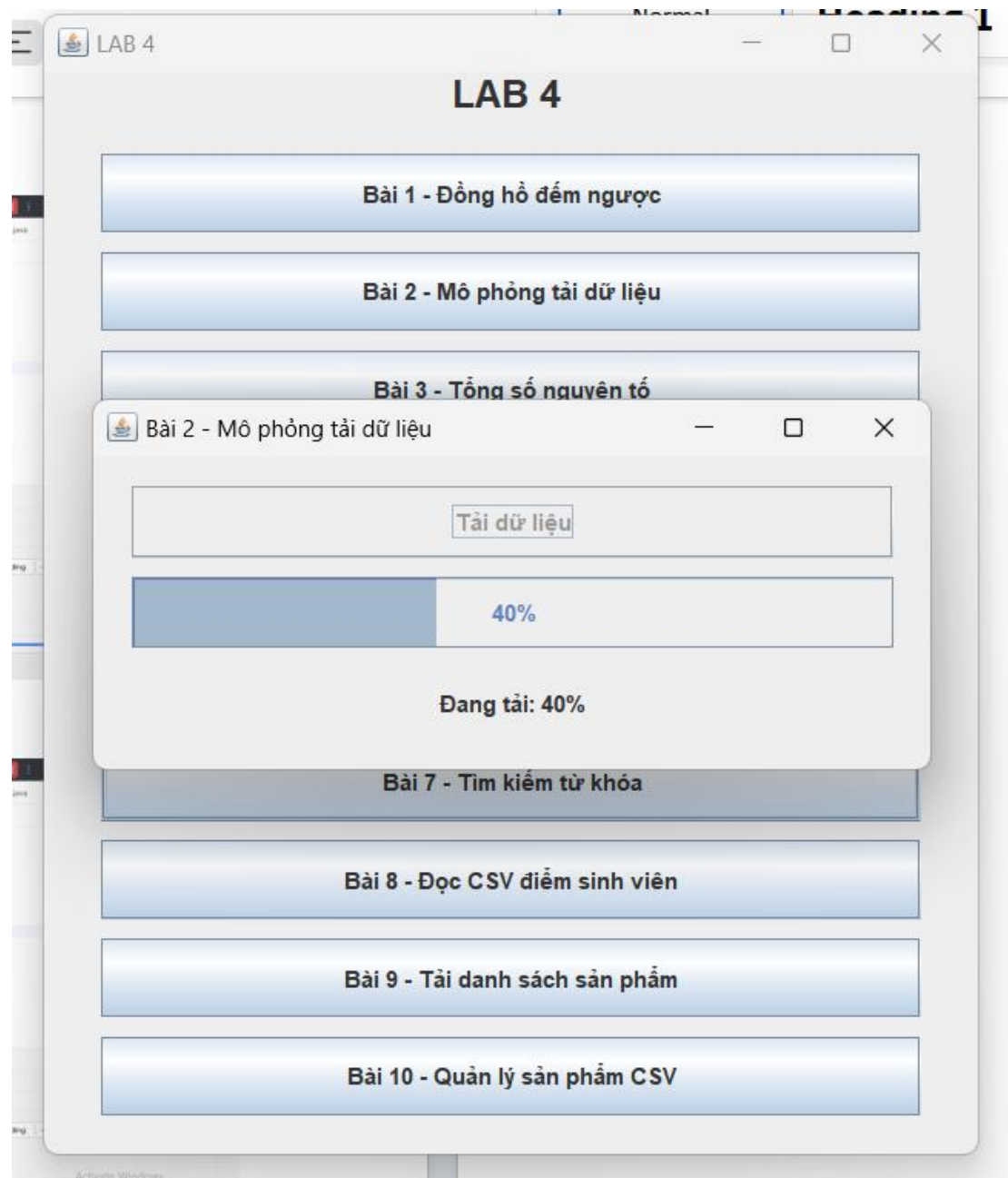
Kết quả đã đạt được



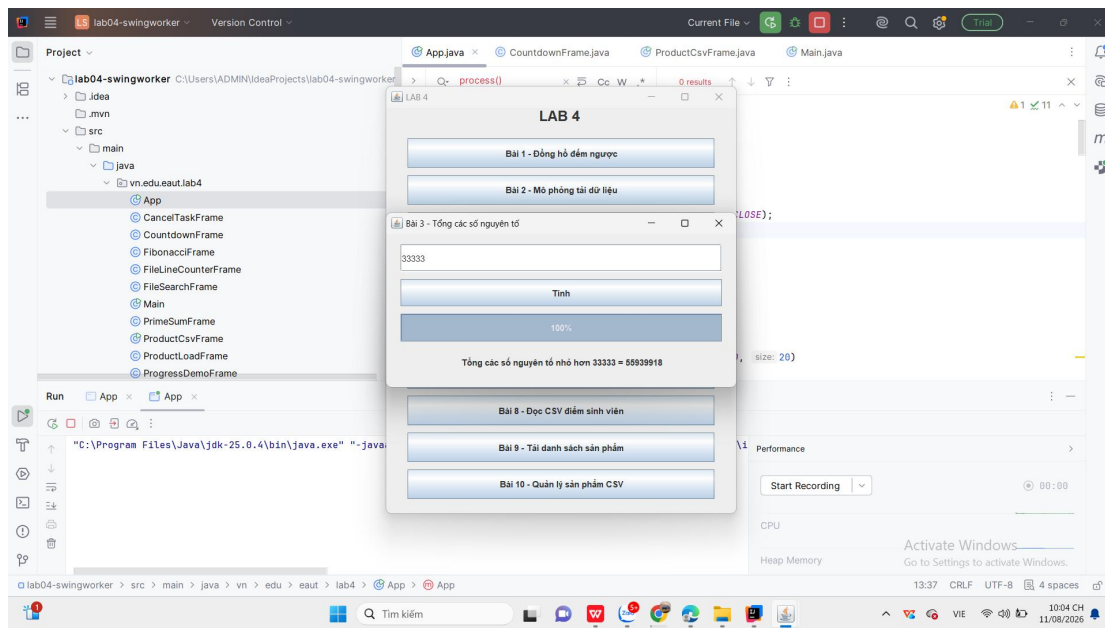
Bài 1



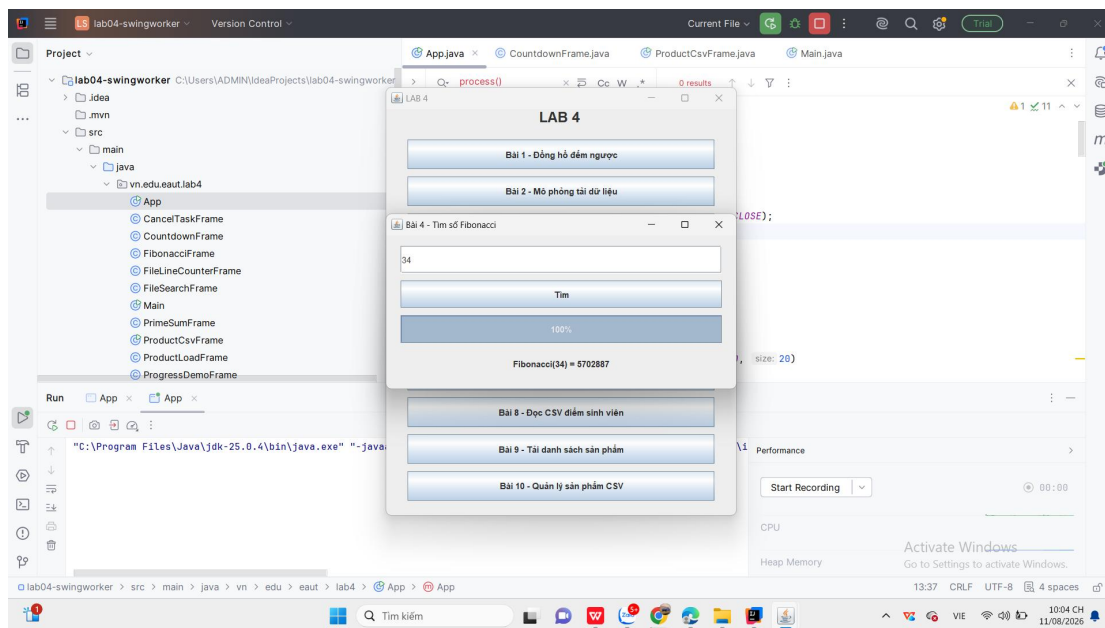
Bài 2



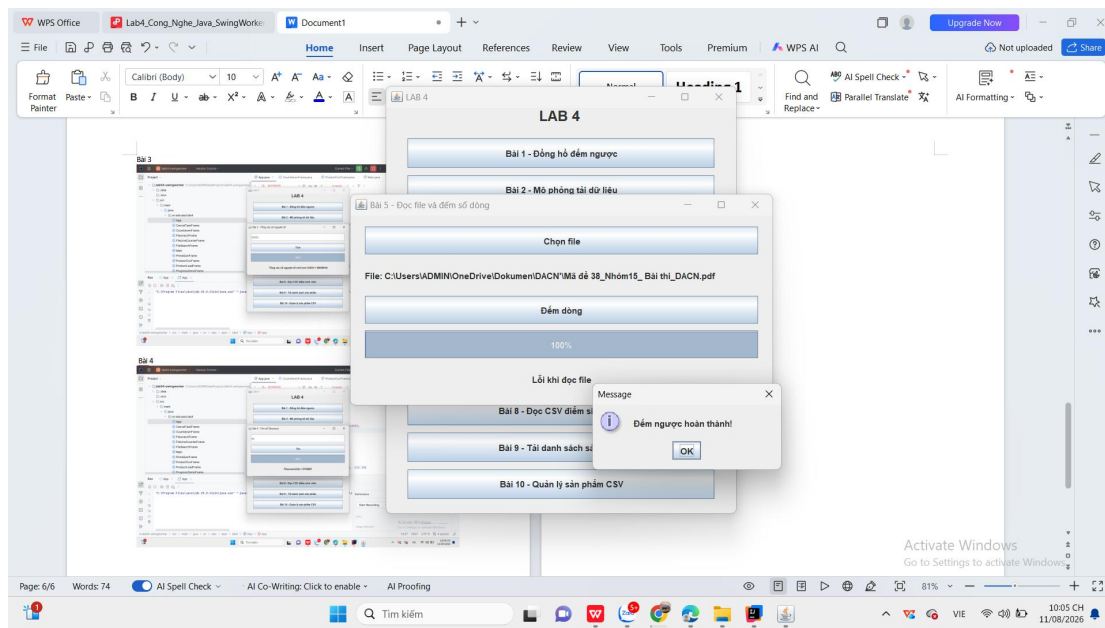
Bài 3



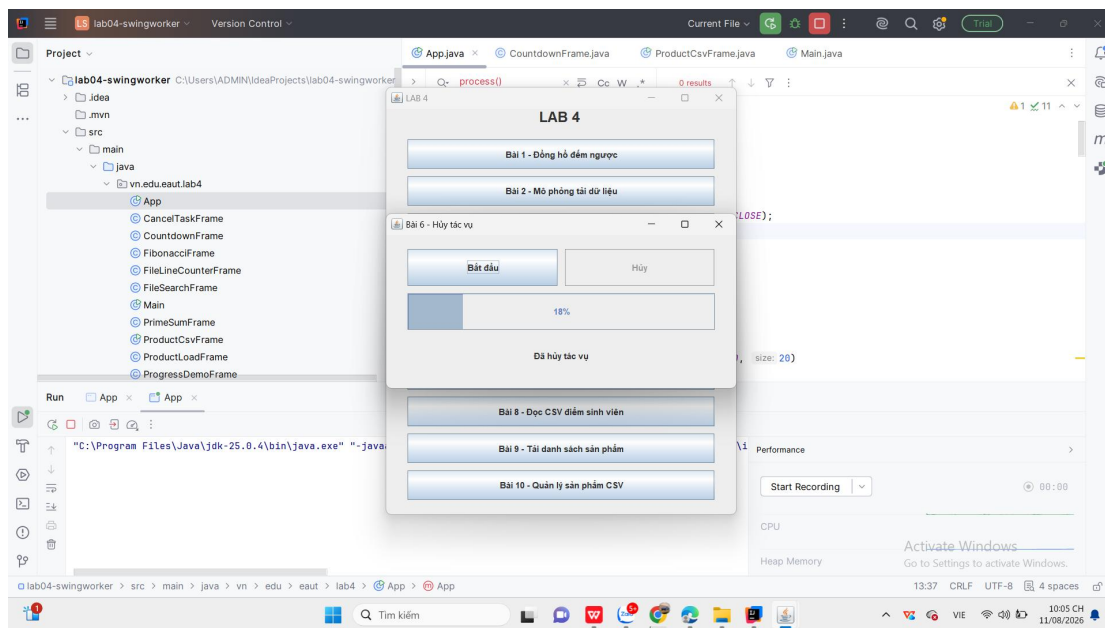
Bài 4



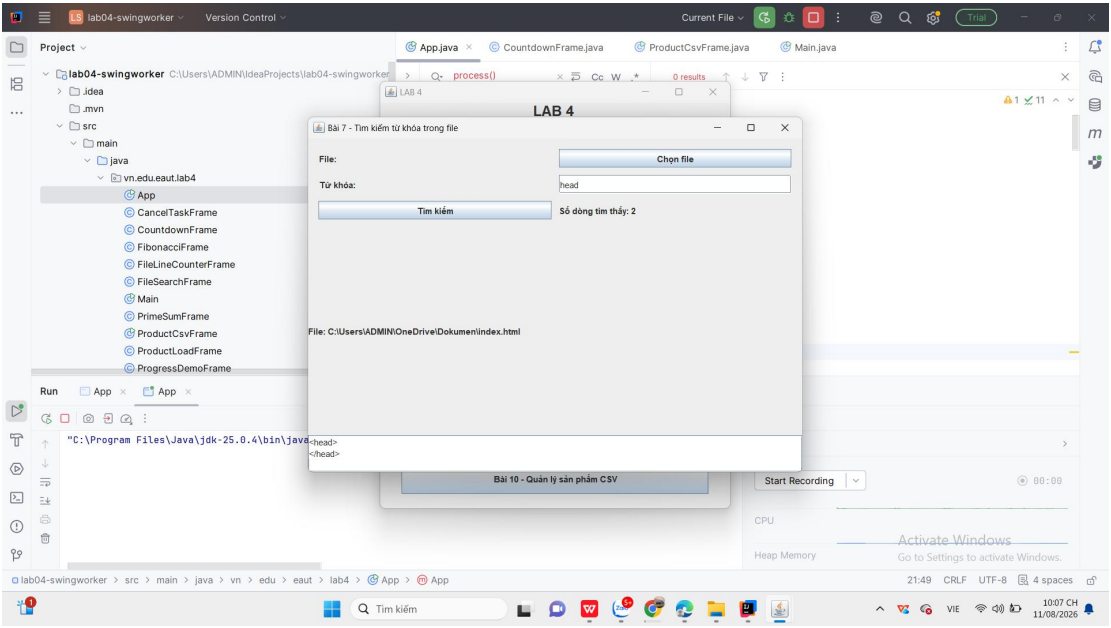
Bài 5



Bài 6



Bài 7



Bài 9

